


# CS421 – Computer Networks

Section 2

Wireshark Assignment

Fall 2025

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## 1) Wireshark Lab: HTTP

### *a) The Basic HTTP GET/response interaction*

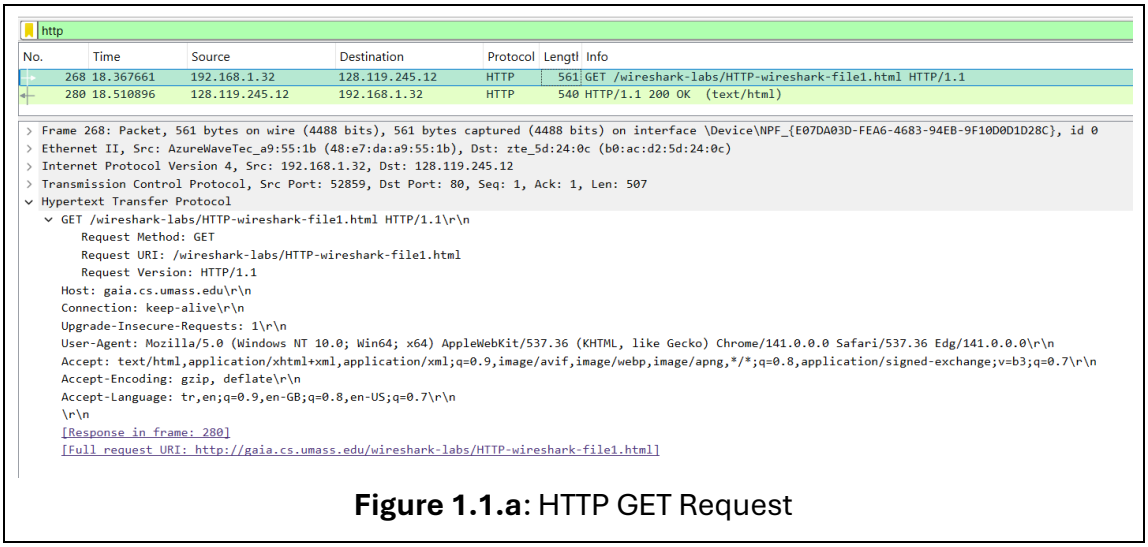



**Figure 1.1.a**: HTTP GET Request

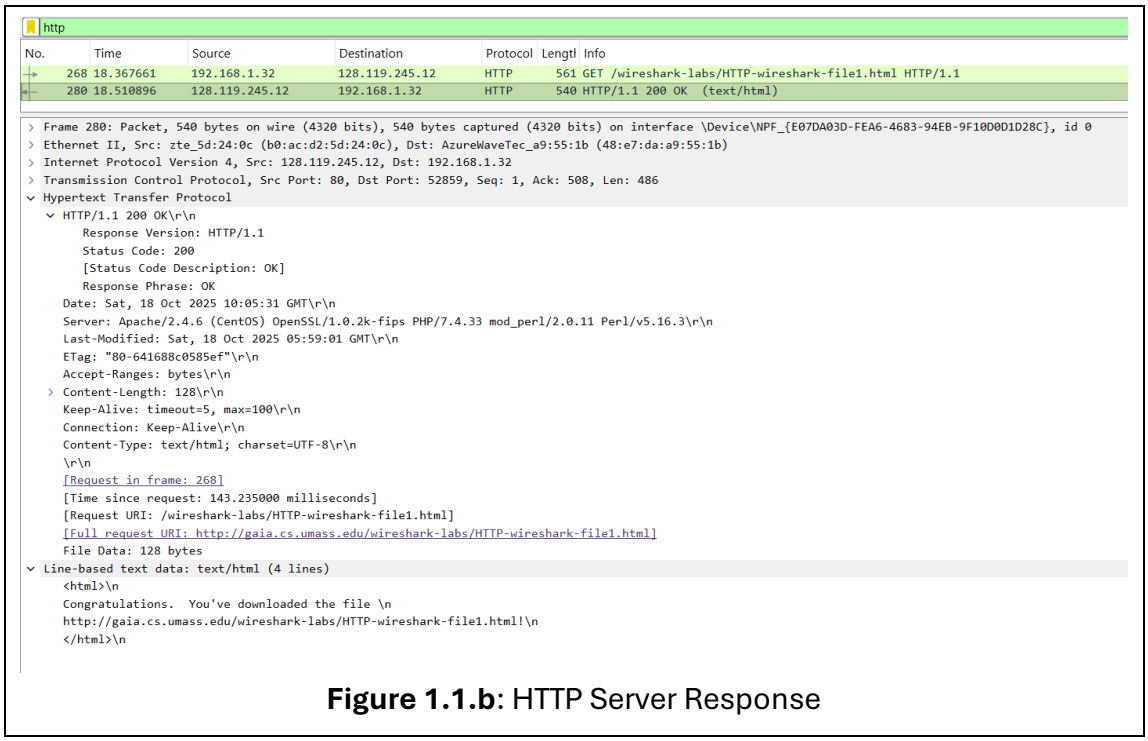



**Figure 1.1.b**: HTTP Server Response

1. Is your browser running HTTP version 1.0, 1.1, or 2? What version of HTTP is the server running?
   - Both the server and the browser are running **HTTP 1.1**. This is shown in the first line of request and response packets (**GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1**, and **HTTP/1.1 200 OK**).

2. What languages (if any) does your browser indicate that it can accept to the server?

- My browser can accept two languages: **Turkish** and **English**. This info can be found in the "Accept-Language" header on the GET request.

3. What is the IP address of your computer? What is the IP address of the gaia.cs.umass.edu server?
   - My Computer: **192.168.1.32**
   - Gaia Server: **128.119.245.12**

4. What is the status code returned from the server to your browser?
   - **200 OK**, meaning the request was succesful.

5. When was the HTML file that you are retrieving last modified at the server?
   - According to the "Last-Modified" header on the response packet, it is **Saturday, October 18, 05.59.01 GMT**.

6. How many bytes of content are being returned to your browser?
   - **128 bytes**, as stated in the "Content-Length" header of request packet.

7. By inspecting the raw data in the packet content window, do you see any headers within the data that are not displayed in the packet-listing window? If so, name one.
   - There are many, like "**Server**", "**Last-Modified**", or "**Content-Length**"

*b) The HTTP CONDITIONAL GET/response interaction*

1. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE" line in the HTTP GET?
   - **No**, the first GET request doesn't have that.

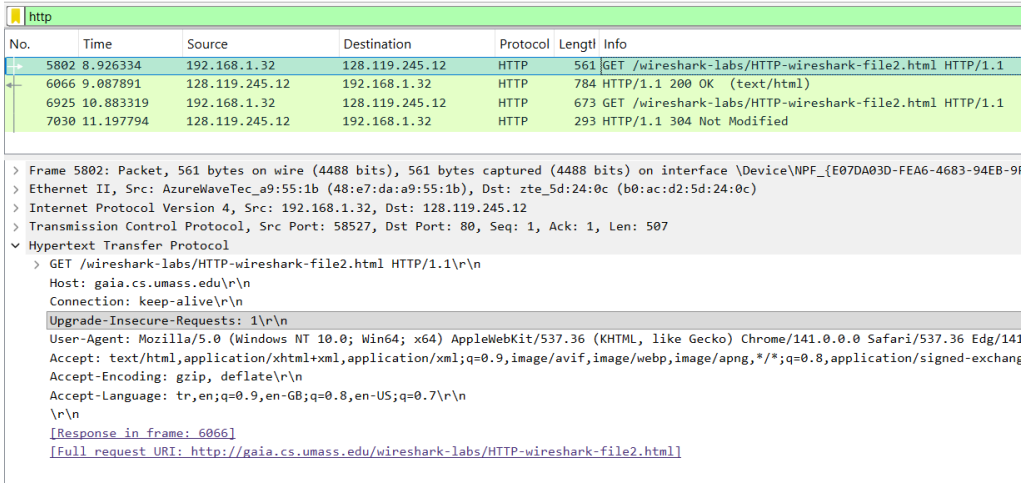



**Figure 1.2.a**: First GET Request

2. Inspect the contents of the server response. Did the server explicitly return the contents of the file? How can you tell?

- For the first request, HTTP response returned with “200 OK” and it returned the contents of the file (see Figure 1.2.b). For the second request, the response returned with “304 Not Modified” and did not return the contents of the file (see Figure 1.2.c). This is observable in the figures that the first response includes a “Line-based text data” part, and the second does not.

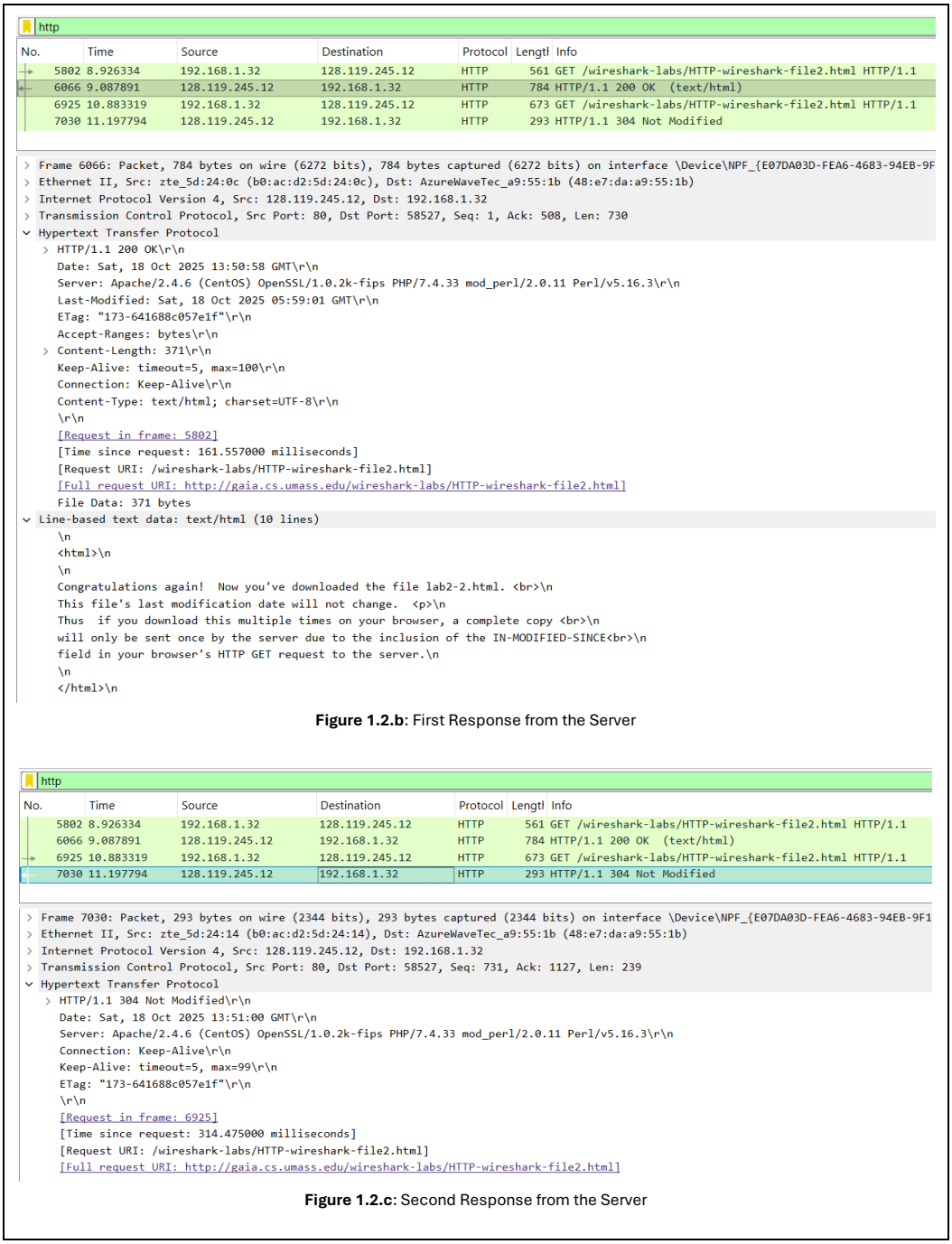



**Figure 1.2.b**: First Response from the Server




**Figure 1.2.c**: Second Response from the Server

3. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET? If so, what information follows the “IF-MODIFIED-SINCE:” header?
   - **Yes**. The second GET request includes this information (see Figure 1.2.d). This is a timestamp that my browser sends to the server the last time it

downloaded the file. The server does not send the content of the file if it is not modified after the browsers download, as it happened in this case.

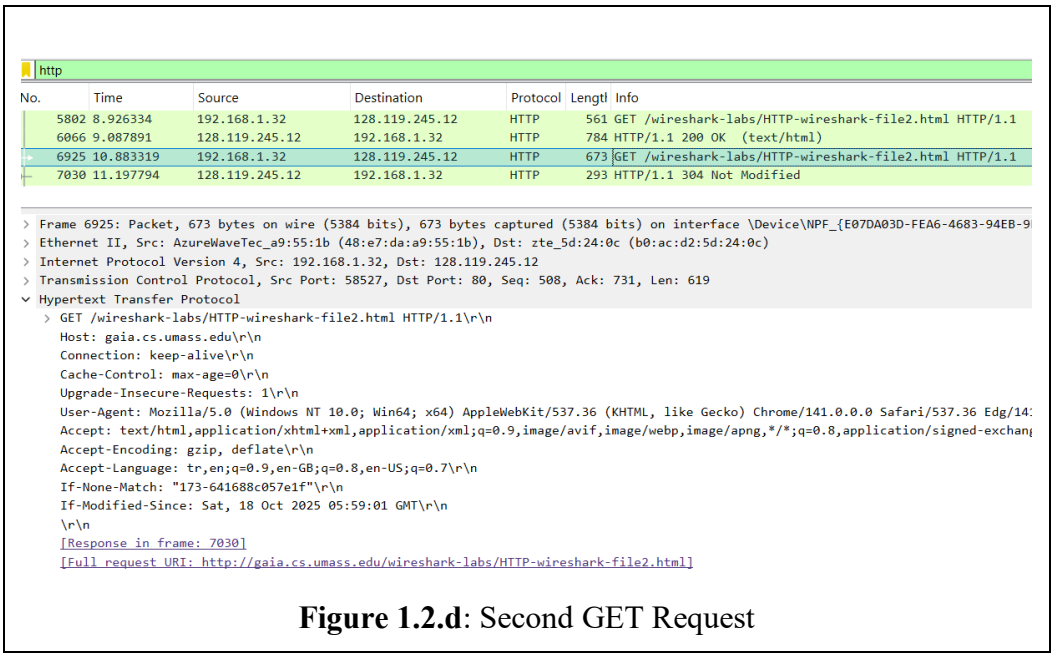



**Figure 1.2.d**: Second GET Request

4. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.
   - As seen in Figure 1.2.c, the second response from the server has the status code **304** with state **Not Modified**. The server did not return the contents of the file since the file is not modified since the time the browser downloaded the file.

*c) Retrieving Long Documents*

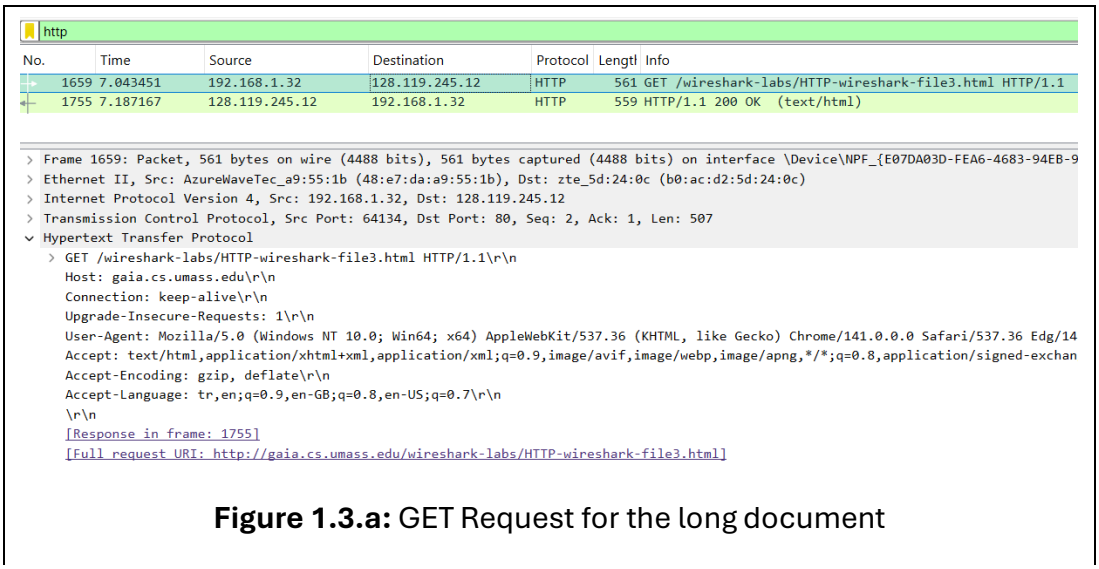



**Figure 1.3.a:** GET Request for the long document

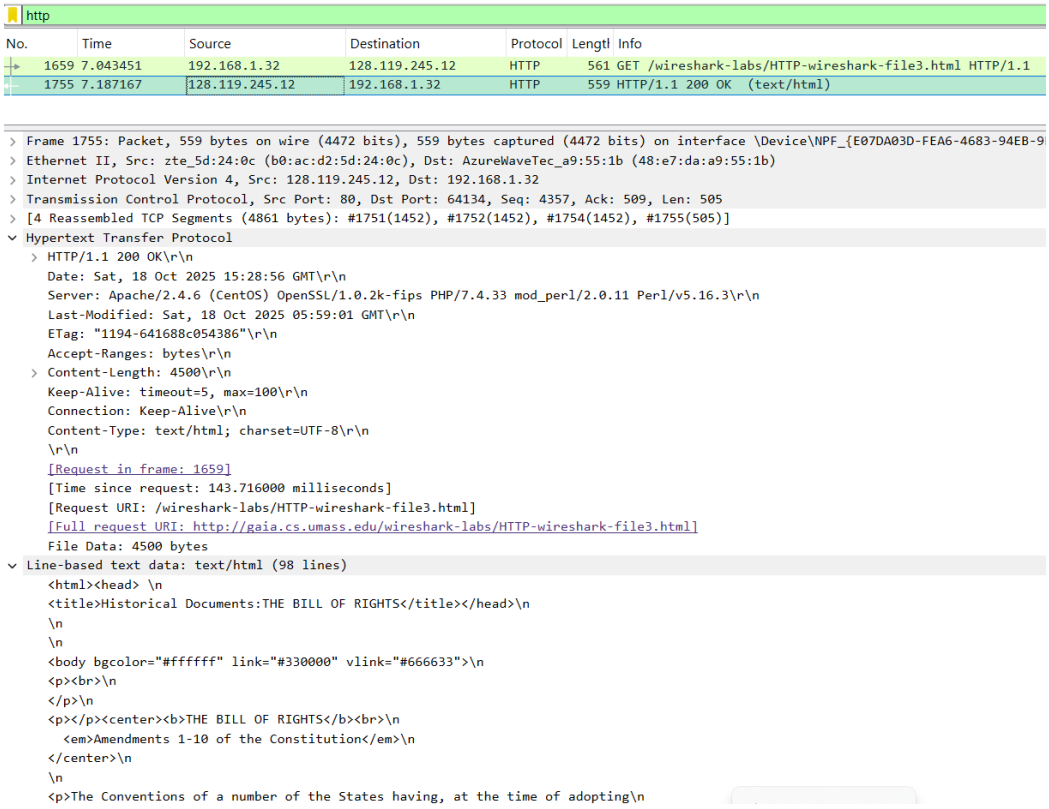



**Figure 1.3.b**: Response for the long document

1. How many HTTP GET request messages did your browser send? Which packet number in the trace contains the GET message for the Bill or Rights?
   - There is **one** GET request message, with packet number **1659** (see Figure 1.3.a).

2. Which packet number in the trace contains the status code and phrase associated with the response to the HTTP GET request?
   - The response with the status code is in packet number **1755** (see Figure 1.3.b).

3. What is the status code and phrase in the response?
   - The status code is **200** and the phrase is **OK** (see Figure 1.3.b).

4. How many data-containing TCP segments were needed to carry the single HTTP response and the text of the Bill of Rights?
   - **Four** TCP segments were needed, as shown in Figure 1.3.b.

*d) HTML Documents with Embedded Objects*

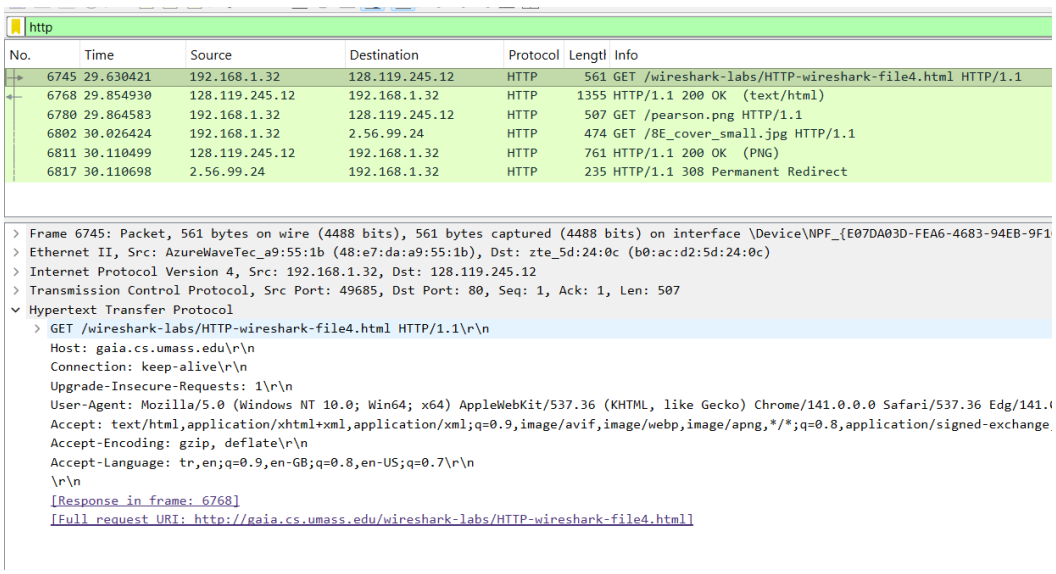



**Figure 1.4.a**: HTTP Document with embedded objects

1. How many HTTP GET request messages did your browser send? To which Internet addresses were these GET requests sent?
   - **Three** GET request messages were sent. One is "pearson.png" from the host "gaia.cs.umass.edu" with IP address "192.168.1.32", and the other is "8E_cover_small.jpg" from the host "kurose.cslash.net" with IP address "2.56.99.24".
2. Can you tell whether your browser downloaded the two images serially, or whether they were downloaded from the two web sites in parallel? Explain.
   - Since the second request was made (30.026424) before receiving the first image (30.110499), the browser should have downloaded the two images in parallel.

*e) HTTP Authentication*

1. What is the server's response (status code and phrase) in response to the initial HTTP GET message from your browser?
   - It responsed with "401 Unauthorized" (see Figure 1.5.a).

2. When your browser's sends the HTTP GET message for the second time, what new field is included in the HTTP GET message?
   - The "Authorization" field is added with "credentials" part (see Figure 1.5.b and 1.5.c).

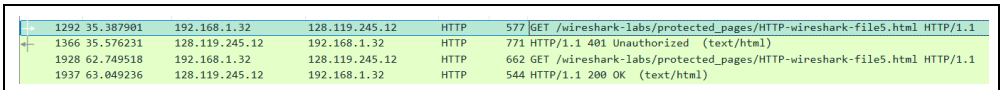
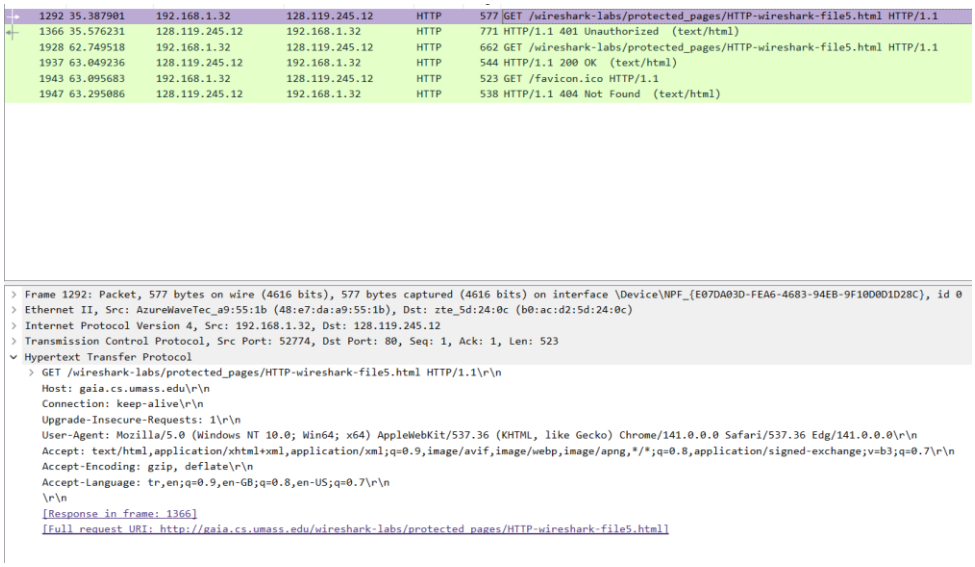


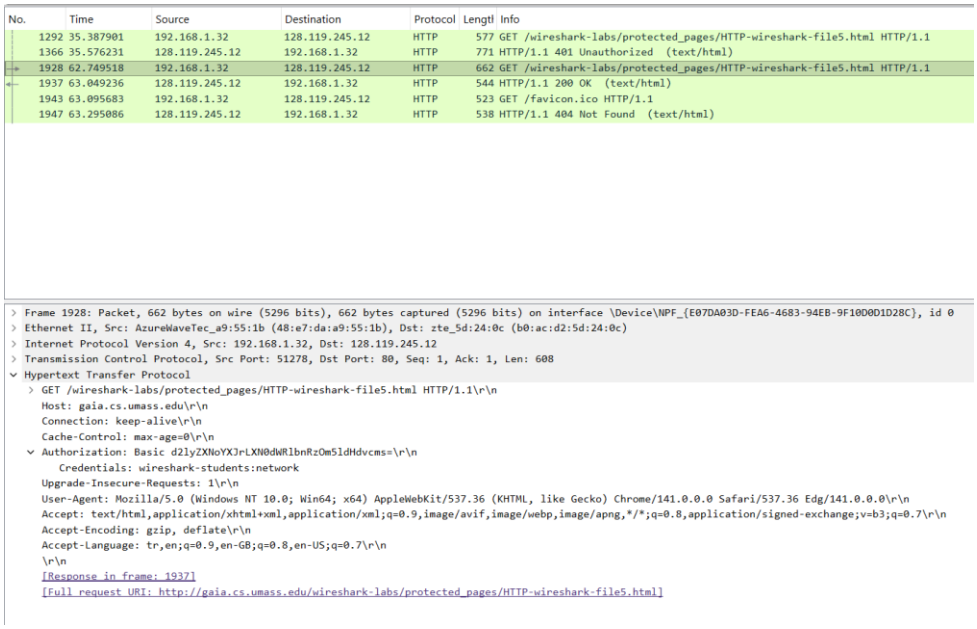

**Figure 1.5.a**: GET Requests and their responses to a password protected site



| | | | | | | |
|---|---|---|---|---|---|---|
| 1292 | 35.387901 | 192.168.1.32 | 128.119.245.12 | HTTP | 577 | GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1 |
| 1366 | 35.576231 | 128.119.245.12 | 192.168.1.32 | HTTP | 771 | HTTP/1.1 401 Unauthorized (text/html) |
| 1928 | 62.749518 | 192.168.1.32 | 128.119.245.12 | HTTP | 662 | GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1 |
| 1937 | 63.049236 | 128.119.245.12 | 192.168.1.32 | HTTP | 544 | HTTP/1.1 200 OK (text/html) |
| 1943 | 63.095683 | 192.168.1.32 | 128.119.245.12 | HTTP | 523 | GET /favicon.ico HTTP/1.1 |
| 1947 | 63.295086 | 128.119.245.12 | 192.168.1.32 | HTTP | 538 | HTTP/1.1 404 Not Found (text/html) |

```
> Frame 1292: Packet, 577 bytes on wire (4616 bits), 577 bytes captured (4616 bits) on interface \Device\NPF_{E07DA03D-FEA6-4683-94EB-9F10D0D1D28C}, id 0
> Ethernet II, Src: AzureWaveTec_a9:55:1b (48:e7:da:a9:55:1b), Dst: zte_5d:24:0c (b0:ac:d2:5d:24:0c)
> Internet Protocol Version 4, Src: 192.168.1.32, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 52774, Dst Port: 80, Seq: 1, Ack: 1, Len: 523
v Hypertext Transfer Protocol
  > GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    Connection: keep-alive\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 Edg/141.0.0.0\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: tr,en;q=0.9,en-GB;q=0.8,en-US;q=0.7\r\n
    \r\n
    [Response in frame: 1366]
    [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/protected_pages/HTTP-wireshark-file5.html]
```

**Figure 1.5.b**: Content of the first GET request



| No. | Time | Source | Destination | Protocol | Lengtl | Info |
|---|---|---|---|---|---|---|
| 1292 | 35.387901 | 192.168.1.32 | 128.119.245.12 | HTTP | 577 | GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1 |
| 1366 | 35.576231 | 128.119.245.12 | 192.168.1.32 | HTTP | 771 | HTTP/1.1 401 Unauthorized (text/html) |
| 1928 | 62.749518 | 192.168.1.32 | 128.119.245.12 | HTTP | 662 | GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1 |
| 1937 | 63.049236 | 128.119.245.12 | 192.168.1.32 | HTTP | 544 | HTTP/1.1 200 OK (text/html) |
| 1943 | 63.095683 | 192.168.1.32 | 128.119.245.12 | HTTP | 523 | GET /favicon.ico HTTP/1.1 |
| 1947 | 63.295086 | 128.119.245.12 | 192.168.1.32 | HTTP | 538 | HTTP/1.1 404 Not Found (text/html) |

```
> Frame 1928: Packet, 662 bytes on wire (5296 bits), 662 bytes captured (5296 bits) on interface \Device\NPF_{E07DA03D-FEA6-4683-94EB-9F10D0D1D28C}, id 0
> Ethernet II, Src: AzureWaveTec_a9:55:1b (48:e7:da:a9:55:1b), Dst: zte_5d:24:0c (b0:ac:d2:5d:24:0c)
> Internet Protocol Version 4, Src: 192.168.1.32, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 51278, Dst Port: 80, Seq: 1, Ack: 1, Len: 608
v Hypertext Transfer Protocol
  > GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    Connection: keep-alive\r\n
    Cache-Control: max-age=0\r\n
  v Authorization: Basic d2lyZXNoYXJrLXN0dWRlbnRzOm5ldHdvcms=\r\n
      Credentials: wireshark-students:network
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 Edg/141.0.0.0\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: tr,en;q=0.9,en-GB;q=0.8,en-US;q=0.7\r\n
    \r\n
    [Response in frame: 1937]
    [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/protected_pages/HTTP-wireshark-file5.html]
```

**Figure 1.5.c**: Content of the second GET request

## 2) Wireshark Lab: DNS

```
C:\Users\ASUS>nslookup www.iitb.ac.in
Server:  csp3.zte.com.cn
Address:  192.168.1.1

Non-authoritative answer:
Name:    www.iitb.ac.in
Address:  103.21.124.133
```

**Figure 2.1.a**: IPv4 Address of www.iitb.ac.in

*a) Nslookup*

1. Run nslookup to obtain the IP address of the web server for the Indian Institute of Technology in Bombay, India: www.iitb.ac.in. What is the IP address of www.iitb.ac.in
   - It is **103.21.124.133** (see Figure 2.1.a).

2. What is the IP address of the DNS server that provided the answer to your nslookup command in question 1 above?
   - It is **192.168.1.1** (see Figure 2.1.a).

3. Did the answer to your nslookup command in question 1 above come from an authoritative or non-authoritative server?
   - It came from a **non-authoritative server** (see Figure 2.1.a), meaning the provider DNS server do not have the ownership of the specific domain (so the IP information was cached).

4. Use the nslookup command to determine the name of the authoritative name server for the iit.ac.in domain. What is that name? (If there are more than one authoritative servers, what is the name of the first authoritative server returned by nslookup)? If you had to find the IP address of that authoritative name server, how would you do so?
   - The first authoritative server is **dns3.iitb.ac.in** (see Figure 2.1.b). To find its IP address, I used "nslookup dns3.iitb.ac.in" command as in Figure 2.1.c, and it is **103.21.127.129**.

```
C:\Users\ASUS>nslookup -type=NS iitb.ac.in
Server:  csp1.zte.com.cn
Address:  192.168.1.1

Non-authoritative answer:
iitb.ac.in      nameserver = dns3.iitb.ac.in
iitb.ac.in      nameserver = dns2.iitb.ac.in
iitb.ac.in      nameserver = dns1.iitb.ac.in
```

**Figure 2.1.b**: Authoritative Name Servers of www.iitb.ac.in

```
C:\Users\ASUS>nslookup dns3.iitb.ac.in
Server:  csp3.zte.com.cn
Address:  192.168.1.1

Non-authoritative answer:
Name:    dns3.iitb.ac.in
Address:  103.21.127.129
```

**Figure 2.1.c**: IP Address of dns3.iitb.ac.in

*b) The DNS cache on your computer*

No questions in this part.

*c) Tracing DNS with Wireshark*

1. Locate the first DNS query message resolving the name gaia.cs.umass.edu. What is the packet number in the trace for the DNS query message? Is this query message sent over UDP or TCP?
   - The packet number is **60**, and it is sent over **UDP** (see Figure 2.3.a).

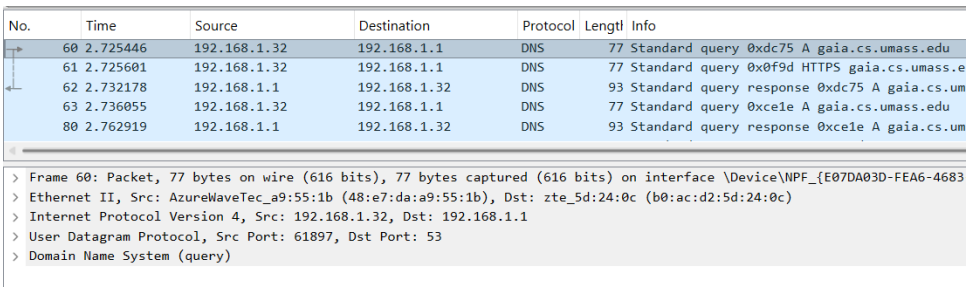



**Figure 2.3.a**: First DNS Query Message

2. Now locate the corresponding DNS response to the initial DNS query. What is the packet number in the trace for the DNS response message? Is this response message received via UDP or TCP?
   - The packet number is **62**, and it is received via **UDP** (see Figure 2.3.b).

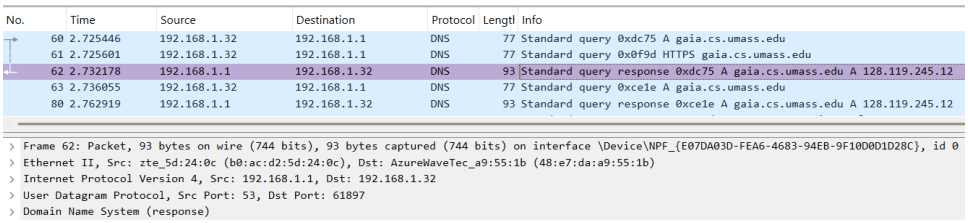



**Figure 2.3.b**: Response for the First DNS Query Message

3. What is the destination port for the DNS query message? What is the source port of the DNS response message?
   - The destination port for the DNS query message is **53** (see Figure 2.3.a). The source port of the DNS response message is also **53** (see Figure 2.3.b).

4. To what IP address is the DNS query message sent?
   - The IP address is sent to **192.168.1.1** (see Figure 2.3.a).

5. Examine the DNS query message. How many "questions" does this DNS message contain? How many "answers" answers does it contain?
   - It contains **1 question** and **0 answers** (see Figure 2.3.c).

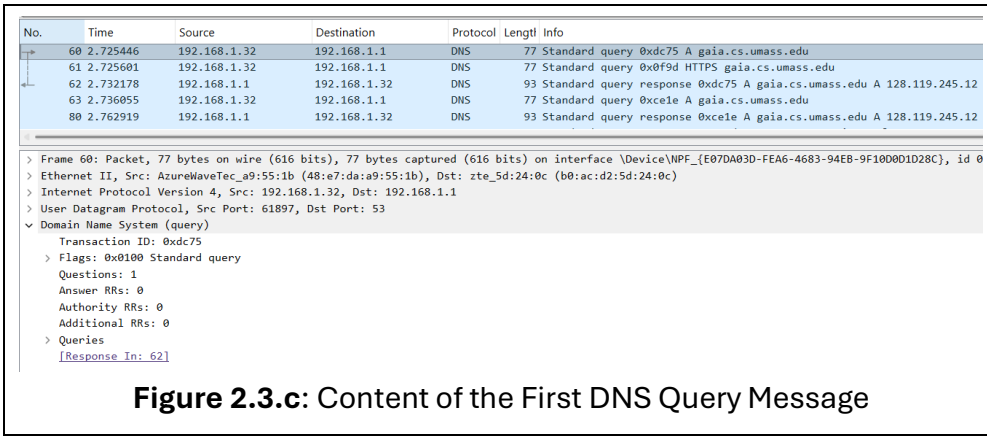



**Figure 2.3.c**: Content of the First DNS Query Message

6. Examine the DNS response message to the initial query message. How many "questions" does this DNS message contain? How many "answers" answers does it contain?
   - It contains **1 question** and **1 answer** (see Figure 2.3.d).

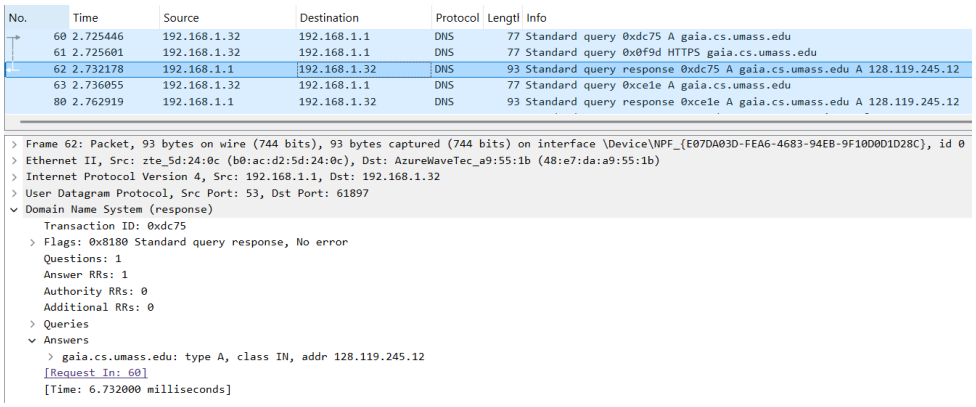



**Figure 2.3.d**: Content of the Response Message of First DNS Query

7. The web page for the base file http://gaia.cs.umass.edu/kurose_ross/ references the image object http://gaia.cs.umass.edu/kurose_ross/header_graphic_book_8E_2.jpg , which, like the base webpage, is on gaia.cs.umass.edu. What is the packet number in the trace for the initial HTTP GET request for the base file http://gaia.cs.umass.edu/kurose_ross/? What is the packet number in the trace of the DNS query made to resolve gaia.cs.umass.edu so that this initial HTTP request can be sent to the gaia.cs.umass.edu IP address? What is the packet number in the trace of the received DNS response? What is the packet number in the trace for the HTTP GET request for the image object http://gaia.cs.umass.edu/kurose_ross/header_graphic_book_8E2.jpg? What is the packet number in the DNS query made to resolve gaia.cs.umass.edu so that this second HTTP request can be sent to the gaia.cs.umass.edu IP address? Discuss how DNS caching affects the answer to this last question.
   - The packet number of HTTP GET request for the base file: **95** (see Figure 2.3.e)
   - The DNS query for gaia.cs.umass.edu: **60** (see Figure 2.3.a).
   - The response to the DNS query: **62** (see Figure 2.3.b).
   - The request for the image file: **266** (see Figure 2.3.e).
   - There isn't a DNS query made for the second HTTP request. This is because of DNS caching. For the first HTTP request, my computer did not know the IP address of the host, so it sent a DNS query to find out. Then it cached this result, which is the reason why there was no need to send another DNS query.

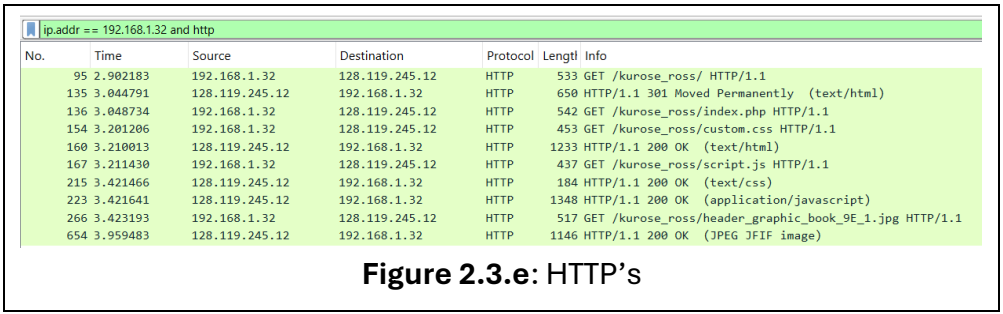



**Figure 2.3.e**: HTTP's

8. What is the destination port for the DNS query message? What is the source port of the DNS response message?
   - The destination port for the DNS query message is **53**, the source port of the DNS response message is the same (see Figure 2.3.e and 2.3.f).

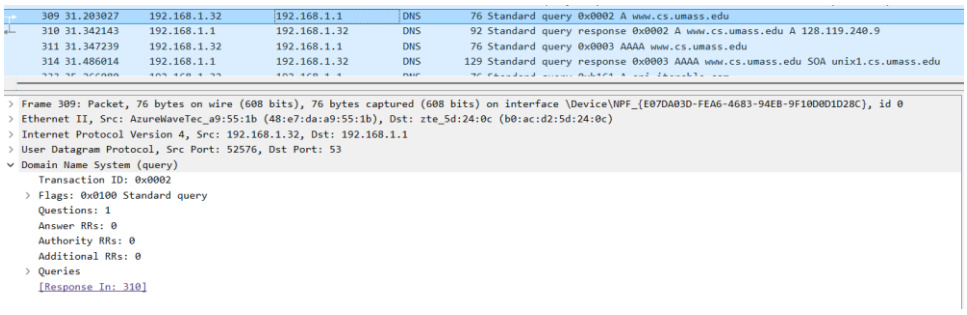



**Figure 2.3.e**: DNS Query Message

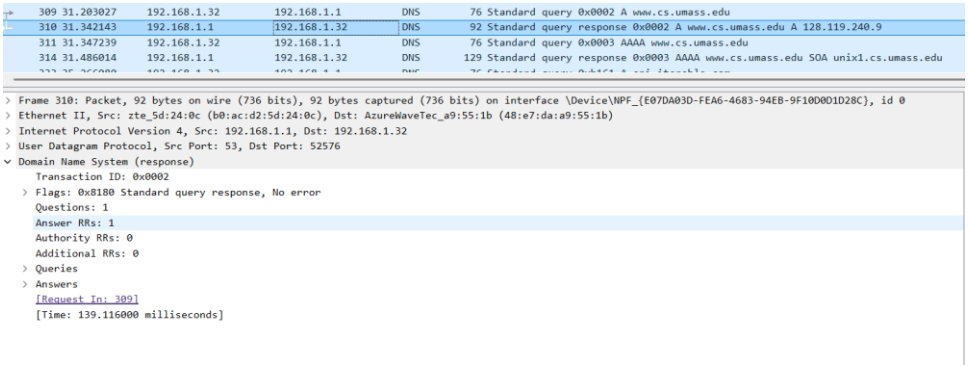



**Figure 2.3.f**: DNS Response Message

9. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?
   - It is sent to the IP address **192.168.1.1** (see Figure 2.3.e). Yes, it is the IP address of my local DNS server, it is not the IP address of the original server.

10. Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"?
    - It is **Type A**, and it does not contain **any answers** (see Figure 2.3.e).

11. Examine the DNS response message to the query message. How many "questions" does this DNS response message contain? How many "answers"?
    - It has **1 question** and **1 answer**.

12. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?
    - It is sent to **192.168.1.1**, and **yes**, it is (see Figure 2.3.g).

```
C:\Users\ASUS>nslookup -type=NS umass.edu
Server:  csp1.zte.com.cn
Address:  192.168.1.1

Non-authoritative answer:
umass.edu       nameserver = ns3.umass.edu
umass.edu       nameserver = ns1.umass.edu
umass.edu       nameserver = ns2.umass.edu
```

**Figure 2.3.g**: Authoratative Name Servers of umass.edu

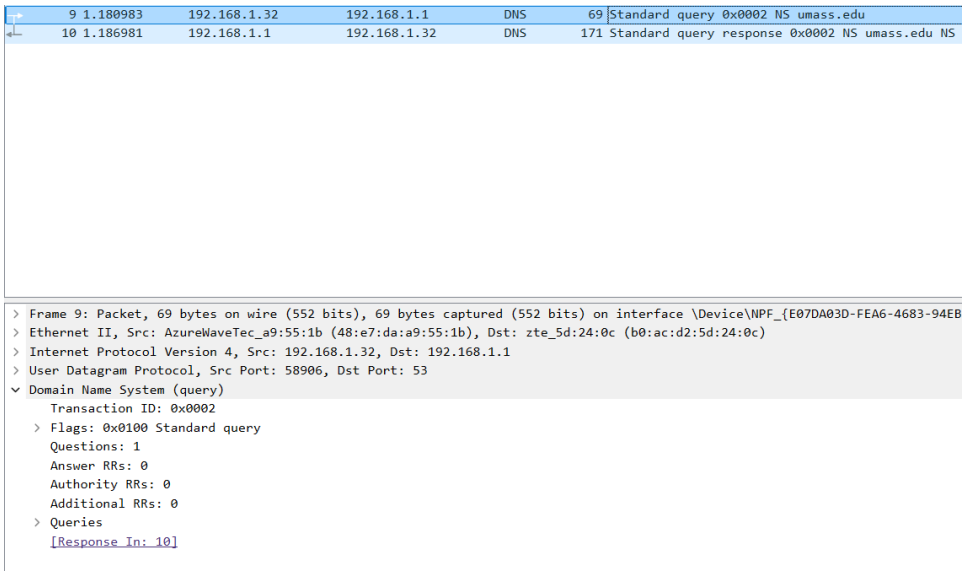



**Figure 2.3.h**: DNS query message on umass.edu

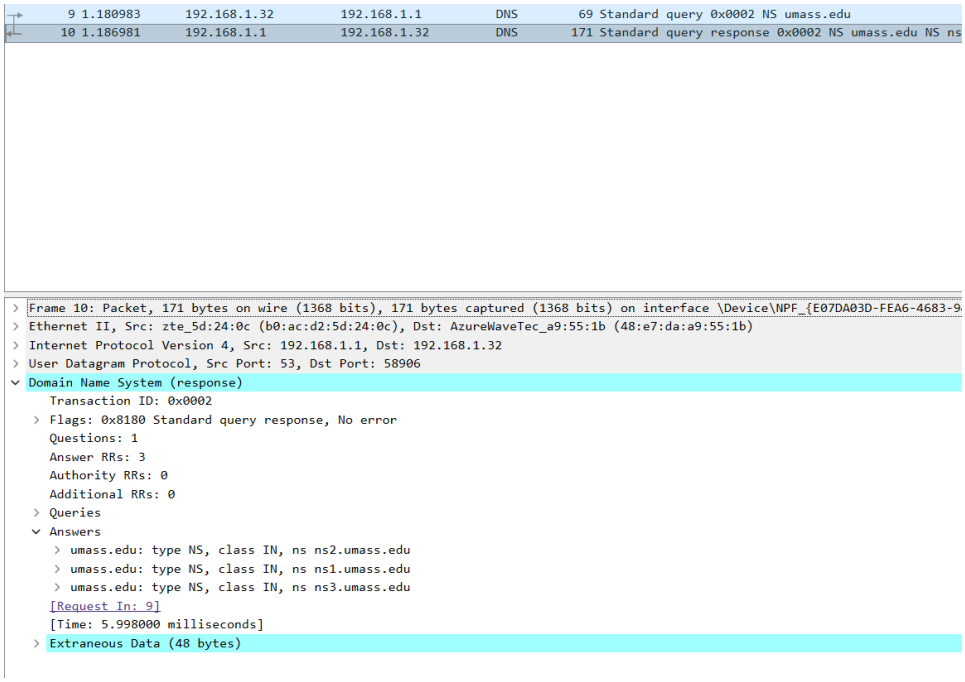



**Figure 2.3.i**: DNS response message of umass.edu

13. Examine the DNS query message. How many questions does the query have? Does the query message contain any “answers”?
    - It contains **1 question** and **0 answers** (see Figure 2.3.h).

14. Examine the DNS response message (in particular the DNS response message that has type “NS”). How many answers does the response have? What information is contained in the answers? How many additional resource records are returned? What additional information is included in these additional resource records (if additional information is returned)?
    - It has **3 answers**. They contain the names of the **authoritative name servers** for the umass.edu domain. **No additional resources are returned**, but appearently it can be expected to see **type A records** in this section.